Athul S

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Summary

M.Tech (Computer Science) candidate specializing in Digital Image Computing, with expertise in machine learning, deep learning, computer vision, and AI applications. Experienced in academic research and projects focused on building intelligent systems to address real-world challenges. Passionate and driven software engineer eager to kick-start a career in the tech industry, bringing fresh perspectives, strong problem-solving abilities, and a commitment to delivering high-quality, impactful software solutions.

Education & Technical Skills

University of Kerala, Master of Technology in Computer Science (2024 – 2026)
Specialization: Digital Image Computing
Coursework: Machine Learning, Deep Learning, Swarm Intelligence, Computer Vision

University College Of Engineering , Kariavattom , TVM, Bachelor of Technology in Information Technology (2019 – 2023)
GPA: 7.36/10
Core Courses: Python, Database Systems, Operating Systems, Data Structures

Sivagiri HSS , Varkala , TVM, Higher Secondary Education (2017 – 2019)
Marks Scored: 90.0%

Jawahar Public Senior Secondary School , Varkala , TVM, High School Education (2016 – 2017)
CGPA: 10/10

Technical Skills

- **Programming:** Python, C, Java
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, OpenCV
- **Data Science:** NumPy , Pandas , Matplotlib , Seaborn , SQL , Pdist , Griddata
- **Optimization:** Swarm Intelligence, Genetic Algorithms, DL Architectures
- **Other:** Odoo, LaTeX, Linux, Jupyter Notebook, Raspberry Pi

Internships

Artificial Intelligence - Machine Learning, Additional Skill Acquisition Programme (ASAP) Kerala Dec 2025 - Jan 2026

- Completed a 4-week internship in Artificial Intelligence Machine Learning using the IBM SkillsBuild platform, gaining hands-on exposure to core AI/ML concepts and tools
- Collaborated in an industry-aligned training program conducted by Edunet Foundation in association with ASAP Kerala, demonstrating active participation and practical learning from Dec 2025 – Jan 2026.

Summer School on Deep Learning , PDPM IIITDM Jabalpur – India May 2025 – Jun 2025

- Attended intensive modules on deep learning theory and practice
- Participated in hands-on workshops on neural networks, optimization, and applications

Python + Artificial Intelligence Intern, Lexyle EdTech , Technopark Phase III, Trivandrum, India Oct 2023 – Feb 2024

- Worked on Python and Artificial Intelligence projects as a part of internship

Certifications

Data Science for Engineers, NPTEL Jan 2025 - Mar 2025

- Successfully completed 8-week NPTEL certification course
- Scored 71/100 (Elite Certification)

Tata Cybersecurity Security Analyst Job Simulation on Forage, Mar 2024

- Completed a job simulation involving identity and access management (IAM) for Tata Consultancy Services,

collaborating with a Cybersecurity Consulting team

- Acquired expertise in IAM principles, cybersecurity best practices, and strategic alignment with business objectives
- Delivered comprehensive documentation and presentations, showcasing the ability to communicate complex technical concepts effectively.

The AI Ladder: A Framework for Deploying AI in your Enterprise Apr 2024

- Completed a course on The AI Ladder: A Framework for Deploying AI in your Enterprise offered by IBM

Fundamentals of Generative AI Nov 2023

- Completed a course on Generative AI offered by Microsoft

Python Programming Mar 2021

- Completed a course on Python covering OOP, Data handling and libraries

Machine Learning Dec 2019

- Participated in a hands-on workshop in Machine Learning jointly organised by University college of Engineering and Department of Computational Biology and Bioinformatics

Projects

CA-Net: Comprehensive Attention Convolutional Neural Networks for Explainable Medical Image Segmentation Sep 2025

- Developed CA-Net, an attention-enhanced CNN integrating spatial, channel, and scale attention to improve medical image segmentation accuracy across varying object positions, shapes, and scales.
- The model provides explainable attention maps for clinical interpretability while achieving efficient and competitive performance with fewer parameters.

Deep Learning Framework for Multi-Class Medical Image Classification Aug 2025

- Developed a deep learning framework for multi-class chest X-ray classification (COVID-19, Pneumonia, Normal) using VGG16 and MobileNetV2, and proposed a hybrid dual-branch fusion model with channel attention to enhance discriminative feature learning.
- The approach achieved improved accuracy, robustness, and interpretability by highlighting disease-relevant lung regions, supporting computer-aided diagnosis.

Machine Learning-Based 3D Segmentation of Mitochondria in Polarized Epithelial Cells Jan 2025

- Developed a machine learning–based 3D segmentation pipeline for mitochondrial networks in polarized epithelial cells using Trainable Weka Segmentation (ImageJ), achieving improved accuracy and efficiency over traditional 2D and threshold-based methods.
- Validated the approach across epithelial cell models (including RPE cells) and performed quantitative mitochondrial morphology analysis, studying the impact of bafilomycin (BAF) treatment.

Computer Vision Based Field Management System Feb 2024

- The system utilizes Optical Character Recognition [OCR] to extract relevant text information from images, eliminating manual data entry. The extracted data undergoes an intelligent sorting algorithm that categorizes and prioritizes data based on predefined criteria like date, keywords or specific data attributes.

Patient Data Management Using Blockchain Aug 2023

- The patient's details pertaining to all healthcare services he/she has received will be managed into an easily accessible format for using anywhere and anytime.

Online Voting System Feb 2023

- Developed an online voting system for my campus mainly by using Java and MySql